KryloCanvas (Web-WhiteBoard)

Created and Designed by Abhimat Chatterjee

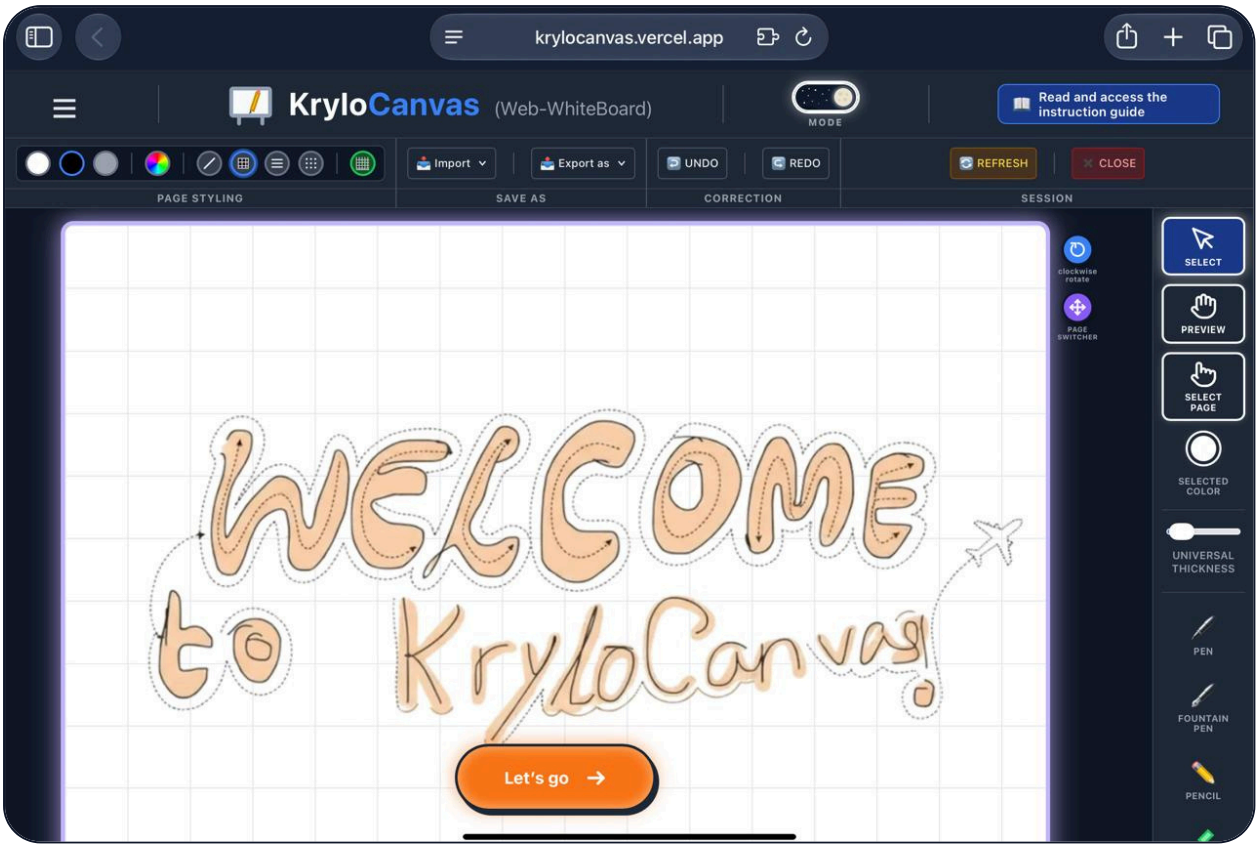


Image: KryloCanvas Start Screen

1. Executive Overview & Architecture

KryloCanvas (Web-WhiteBoard) is a professional, high-precision, browser-based digital whiteboard and mathematical drawing suite for classrooms, STEM coaching institutes and technical educators. It runs inside any modern browser with no installation, supports touch and stylus, and now opens very large PDF, Word, Excel and PowerPoint files without crashing thanks to its lazy-loading page engine.

Core Operational Strengths:

- **Zero Backend:** HTML5 Canvas, Fabric.js, PDF.js and jsPDF in one file.
- **Universal State Engine:** Colour and thickness sync across every tool.
- **Lazy Page Engine:** Huge files open fast, only a few pages awake at a time.

Mathematical & STEM Capabilities:

- **Solid Instruments:** Ruler, Protractor and Set Square with pen snapping.
- **Precision Compass:** Fixed needle, exact radius in cm, mm or inch.
- **43 Shapes & Flowcharts:** Vector shapes, process flows and trees.

QUICK START (6 STEPS)

1. Open **krylocanvas.vercel.app**; on the welcome screen tap **Let's go**.
2. Pick a theme and page style in the top bar; tap **Add Page** for more pages.
3. Choose Pen, Pencil, Highlighter or Text; set colour and thickness.
4. Use Rulers, Compass, Shapes or Flowchart for precise figures.
5. **Import** a PDF, image, Word, Excel or PowerPoint file to annotate it.
6. **Export as** PDF (all or current page) or image; optionally rename.

NEW IN THIS VERSION

- **Lazy loading:** byte-range PDF reading, max 5 live pages, page cache.
- **Word / Excel / PPT import** through the same lazy pipeline.
- **Custom Theme Colour** (spectrum, RGB, HEX); **Select Page** tool.
- **Page Rotate** (90° clockwise) and per-page **Delete**.
- **Rename Export** prefix; **Welcome** start screen.

2. Workspace Layout: Header, Pages & Tool Palette

KryloCanvas has three zones designed for minimum cognitive load while teaching: a **top command header**, a scrolling **pages area** (with a dashed **+ Add Page** button) and a **right-hand tool palette** (a horizontal strip at the bottom on phones).

Top Command Header

- **Instructions** (info icon): in-app help and the QR / link to this guide.
- **Theme dots:** White, Black, Grey and a Custom Theme Colour spectrum button.
- **Page style buttons:** Blank, Standard Grid, Single Lines, Dotted Grid, plus the Millimeter Graph button (5 colours).
- **Import** and **Export as** drop-down menus.
- **Undo / Redo**, **Refresh** and **Close** (both ask for confirmation).

Pages Area & Tool Palette

- Pages are stacked vertically (1200 × 1600 px each) and scroll freely.
- Every page has its own **Rotate** (clockwise) and **Delete** buttons.
- Palette: Select, Preview, Select Page, colour indicator, Pen, Pencil, Highlighter, Text, two Erasers, Shapes, Rulers, Compass, Flowchart, Clear All and the thickness slider.
- Welcome screen on launch: tap **Let's go** to enter the board.

3. Command Bar & Header Reference

Control	Where	What it does
Instructions	Header, left	Opens the help panel and the button / QR to open or download this PDF guide.
White / Black / Grey	Header theme dots	Switches page background; strokes, grid and page stamp invert automatically.
Custom Theme Colour	Header, spectrum dot	Opens colour picker (spectrum, R G B, HEX, saturation, value, contrast, quick shades); changes apply live.
Blank / Grid / Lines / Dots	Header page styles	Changes the paper pattern of the active page (grid, ruled lines or dotted matrix).
Millimeter Graph	Header, graph dot	Adds a new graph-paper page; choose Green, Blue, Red, Orange or B&W.
Undo / Redo	Header	30 steps per page; works on the page you last touched.
Refresh / Close	Header, right	Reload or close the session only after a confirmation dialog.
+ Add Page	End of pages	Appends a new blank page in the chosen theme and updates all page counters.
Rotate / Delete	On each page	Turns that page 90° clockwise with its drawings; or removes the page.

4. Import & Export Menus

Import (file menu)

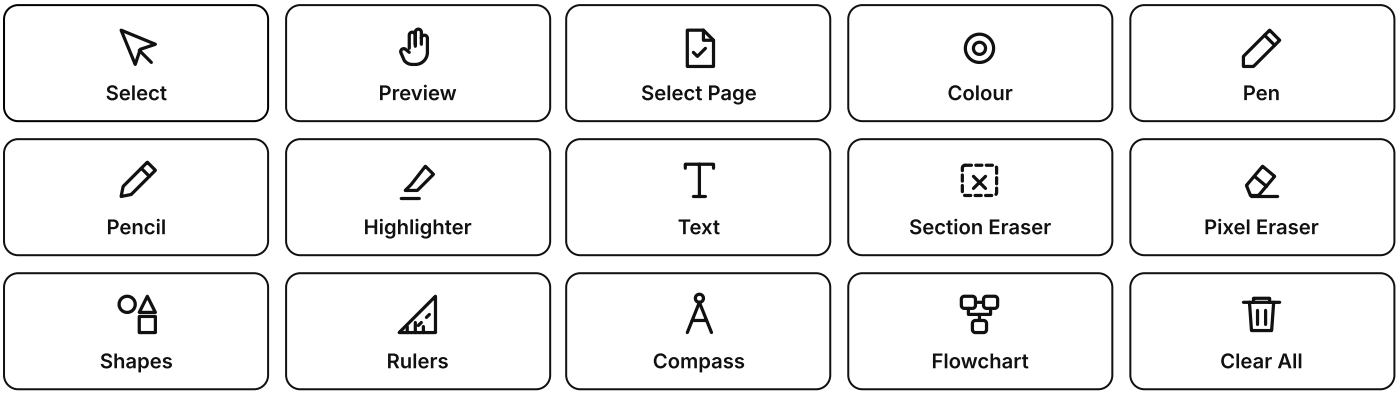
- **PDF** – every page becomes an editable whiteboard page (lazy loaded).
- **Image** – PNG / JPG placed on the page; move, scale, rotate, delete.
- **Word** – .docx (MS Word or Google Docs download), auto-paginated.
- **Excel** – .xlsx, .xls, .csv; 34 rows × 7 columns per page.
- **PowerPoint** – .pptx slides rendered as pages.

Export as

- **PDF (Entire Document)** – all pages, one at a time, to protect memory.
- **PDF (Current Page)** – only the active page.
- **Image (Current Page)** – PNG of the active page; rulers and compass are hidden.
- **Rename Export** dialog: type an optional name; spaces become underscores. Press **Export**, or **Default name** to skip.

5. The Tool Palette at a Glance

The palette on the right lists every tool in a fixed order. Tap a tool once to activate it; the active tool is highlighted with a frame. The universal **thickness slider** (1–40) sits at the bottom.



6. Navigation: Select, Preview & Select Page

Select Tool

- Tap any object to get its bounding box.
- Move, scale, rotate (top handle) or delete drawings, text, shapes, images and flowchart blocks.
- Drag on empty space to marquee-select a group.
- Instruments and images show a **Delete** (top-right) and **Rotate** (top-centre) button.

Preview Tool (Hand)

- Locks drawing so touches only move the view.
- Pinch in / out, or **Ctrl + scroll** on a computer, to zoom from 50 % to 400 %.
- Drag to pan across the pages.
- Pinching on the top or side menus zooms only the menus.

Select Page Tool

- Tap a page to make it the **active page**.
- Drawing is disabled while it is active, so nothing is inked by accident.
- Active page decides where shapes, instruments and Undo / Redo apply.
- Use it before inserting a figure onto a particular page.

7. Selected Colour Popup & Thickness

The round **colour indicator** in the palette shows the live stroke colour. Tapping it opens the Selected Colour popup:

Quick Swatches & Custom Colour

- **10 quick swatches:** Black, Red, Blue, Green, Yellow, Orange, Purple, Pink, Cyan, Grey.
- **Custom colour:** colour wheel / spectrum, saturation and value (HSV), RGB values and a HEX field.
- **Eyedropper:** pick any colour from the screen.
- **Recent colours** are remembered for fast reuse.

Universal Thickness (1–40)

- One slider drives the Pen, Pencil, Highlighter, Shapes and Compass output.
- **Pen:** width = thickness. **Pencil:** half of thickness (minimum 1).
- **Highlighter:** 4 × thickness at 40 % opacity so writing stays visible underneath.
- Change colour or thickness at any time; selected objects can be restyled with Select.

Why the engine is called “Universal”

One shared state holds the colour, thickness and active tool. Switching from Pen to Shapes or Compass never resets your style, so a teacher can keep a chosen colour for a whole lesson, and a single change updates what the next tool draws.

8. Pen, Pencil & Highlighter

Pen (Natural Ink)

- Continuous bold vector strokes in the selected colour.
- Best for stylus writing, derivations and headings.
- `width = thickness`

Pencil (Technical)

- Fine graphite-style line.
- Ideal for construction lines, geometry sketches and diagrams.
- `width = max(1, thickness / 2)`

Highlighter

- Translucent wash that never hides what is below.
- Works over text, formulas and imported pages.
- `width = 4 × thickness, opacity 40 %`

Drawing near a ruler, protractor, set square or compass with any of these three tools **snaps** the stroke to the instrument edge (see Section 16). Every stroke is stored as a vector object, so it can be selected, moved, resized and exported without blur.

9. Eraser Suite: Section vs Pixel Eraser

Eraser	How you use it	Behaviour	Best for
Section Eraser	Drag a dashed rectangular box across the page.	On release, every object enclosed by the box is deleted in one step (one Undo restores it).	Clearing whole proofs, diagrams or notes.
Pixel (Scribble) Eraser	Scribble over the line to be removed.	Round brush erases the part of the stroke it touches and leaves the rest.	Fixing a minus sign, an exponent or one small line.

10. Text Tool

Writing Text

- Choose **Text**, tap anywhere on a page and type plain text.
- It takes the current colour; **Enter** starts a new line.
- Tap existing text with the Text tool to edit it.
- An empty text box is removed automatically when you tap away.

Moving & Resizing Text

- Switch to **Select** to move, scale or rotate text.
- Text stays vector, so it prints crisply in the exported PDF.
- Delete selected text from the Select tool controls.

11. Clear All, Undo & Redo

Clear All Drawings

- The red trash button at the bottom of the palette wipes the drawings of the current page.
- Page background, theme and header stamp stay; use Undo straight after if pressed by mistake.

30-Step History per Page

- Each page owns its own Undo / Redo stack of 30 steps.
- The header buttons act on the page you last drew on.
- Importing or deleting pages does not disturb other pages' history.

Pro tips for neat handwriting

- Use **Preview** to zoom to 200 % before writing small formulas.
- Rest your palm freely in Preview or Select Page; no stray ink.
- Write with Pencil, then highlight key lines with the Highlighter.
- Use Section Eraser for big clean-ups and Pixel Eraser for fine edits.

12. Indelible Header Stamp: Date, Day & Page Counter

Every page carries a two-line stamp at the top-right corner so printed exports stay in order:

- **Line 1:** date and weekday, e.g. `07/11/2026 - Saturday`, read from the device clock.
- **Line 2:** page number over total pages, e.g. `4 / 10`.
- Counters renumber automatically when pages are added or deleted.
- Stamp colour inverts with the theme: dark on light pages and light on dark pages.

07/11/2026 - Saturday
4 / 10

13. Themes & Custom Theme Colour

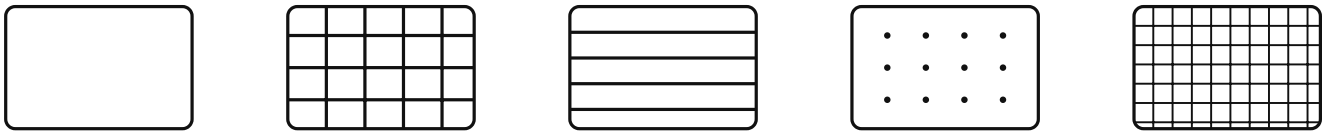
White, Black & Grey

- **White:** clean page for print-ready work.
- **Black:** dark page for projectors; black strokes flip to white automatically.
- **Grey:** neutral studio background.
- Switching theme recolours grid, lines, dots and the stamp.

Custom Theme Colour

- Open the spectrum dot to pick **any** page colour.
- Controls: colour spectrum, white-to-black brightness, R / G / B, HEX, saturation %, value %, contrast and quick shades.
- Applies live while picking.
- Grid, lines and dots become a light tint on dark pages and a dark shade on light pages.

14. Page Styles & 5 Millimeter Graph Papers



Blank | Standard Grid | Single Lines | Dotted Grid | Millimeter

Pick a page style in the header to restyle the active page. The **Millimeter Graph** button adds a *new* page of engineering graph paper in one of five colours: **Green, Blue, Red, Orange** or **Black & White** (high-contrast, ideal for laser printing). Colours invert automatically with the theme.

15. Multi-Page Sessions: Add, Rotate, Delete & Safety

Add & Number

- **+ Add Page** appends a page; Millimeter Graph appends graph pages.
- Pages are 1200 × 1600 px, portrait.
- The stamp “n / total” updates on every page.

Rotate & Delete

- The round arrow on a page turns it **90° clockwise**; its drawings turn with it. Press again for 180°, 270°.
- The bin button removes just that page.

Safety Dialogs

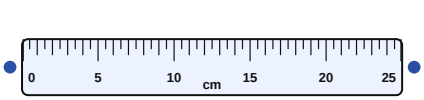
- **Refresh** and **Close** ask before acting.
- Export never deletes your work.
- Undo history is per page and capped at 30 steps.

✓ Recommended classroom flow

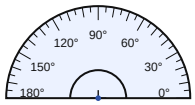
Choose theme and paper → add pages for each topic → write with Pen / Pencil → add figures with Shapes, Rulers and Compass → export **PDF (Entire Document)** and share. The date stamp and “n / total” make every handout traceable.

16. Solid Instruments & Snapping Engine

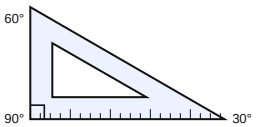
Open **Rulers** in the palette and choose **Ruler**, **Protractor** or **Set Square**. Each instrument appears once per page as a solid, movable object.



Ruler: stretch from the end handles



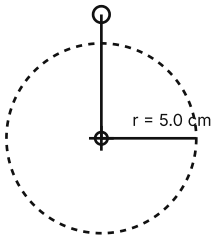
Protractor 0°–180°



Set Square

Topic	How it works
Solid edges	Instruments act as real boundaries: trace with Pen, Pencil or Highlighter and the stroke snaps to the edge, so lines and arcs are perfectly straight or curved.
Snapping range	The pen snaps when it is within 100 px of an instrument edge; keep your stylus near the edge.
Scale	1 cm = 37.8 px and 1 inch = 96 px, matching the on-screen ruler marks.
Ruler	Resize with the left / right handles; its marks regenerate crisply at the new length.
Protractor & Set Square	Move, rotate and delete with the on-object controls; trace for angles, perpendiculars and right-angle triangles.
Controls	Select an instrument: red **Delete** at the top-right, blue **Rotate** at the top-centre. Instruments are hidden in image exports.

17. Precision Compass System



- **Fixed needle:** the red crosshair marks the pivot; drag the compass with **Select** to place it anywhere on any page.
- **Radius:** 0.5 to 500 cm in 0.5 cm steps, with a numeric box for an exact value and a cm / mm / inch readout.
- **Circle:** draws a full 360° circle at the current radius.
- **Arc (90°):** draws an exact 90° arc along the compass arm; rotate the arm 15° at a time.
- **Snapping:** with Pen, Pencil or Highlighter near the compass, strokes lock on to the exact circumference.

HUD control	Action
− 0.5 cm / + 0.5 cm	Shrinks or enlarges the radius by half a centimetre.
Exact radius box	Type any value from 0.5 to 500 and choose the unit.
⟲ 15° / ⟳ 15°	Rotates the compass arm 15° counter-clockwise or clockwise.
Circle / Arc (90°)	Stamps a circle or arc using the universal colour and thickness.
Close (×)	Removes the compass from the page.

18. Classical Constructions with the Compass & Ruler

Because the radius is exact and the pen snaps to the arc, standard constructions can be drawn quickly. Use **Pencil** for construction arcs and **Pen** for the final figure.

Bisect a Line Segment

- Draw segment AB with the Ruler.
- Set a radius larger than half of AB.
- Place the needle on A and draw an arc; repeat from B.
- Join the two crossing points: this is the perpendicular bisector.

Equilateral Triangle

- Draw base AB with the Ruler.
- Set the radius equal to AB.
- Draw an arc from A and another from B.
- Join the crossing point C to A and B.

Bisect an Angle

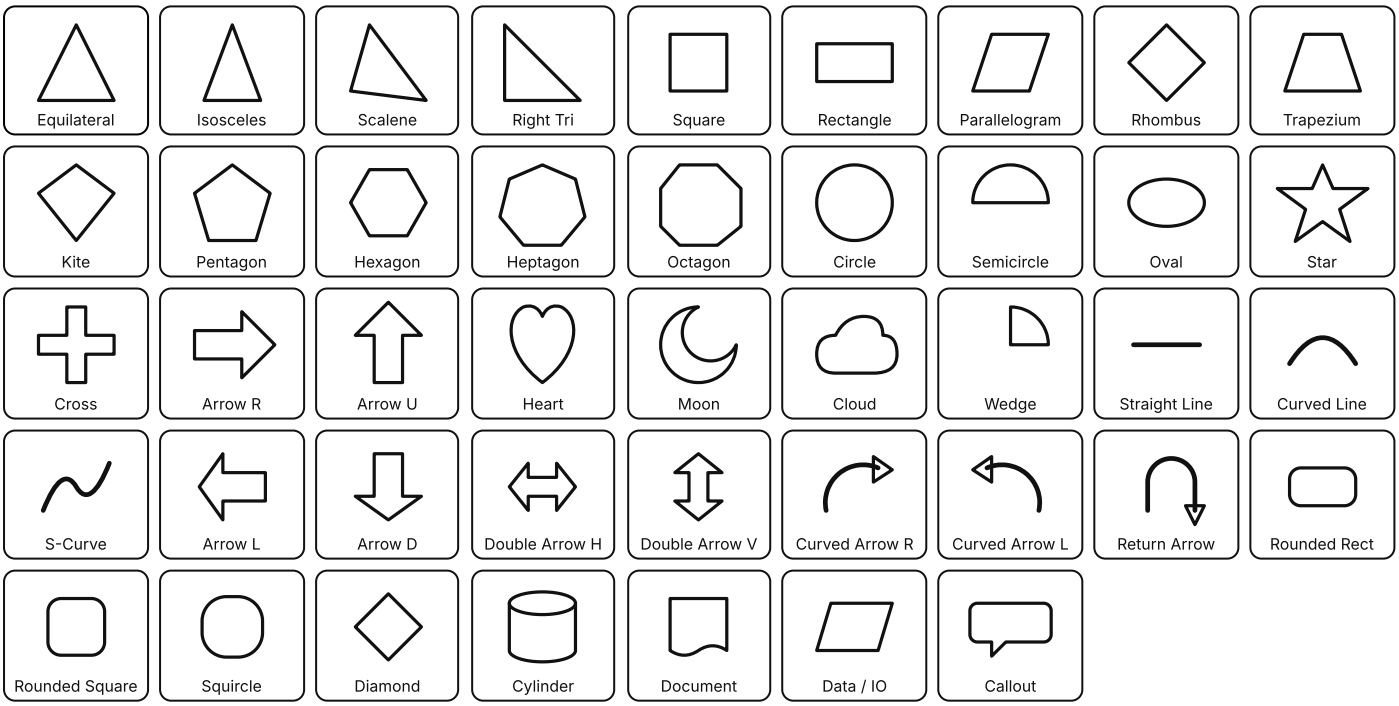
- Draw an arc from the vertex cutting both arms.
- From each cut point draw an arc of equal radius.
- Join the vertex to where the arcs cross.

Regular Hexagon in a Circle

- Draw a Circle of radius r.
- Without changing r, step round the circle with Arc cuts.
- Join six points with the Ruler for a hexagon.

19. The Extended 43-Shape Library

Tap **Shapes** to open the library, then tap a shape to drop it on the active page in the current colour. Shapes use the same universal thickness as ink and remain fully editable.



Groups

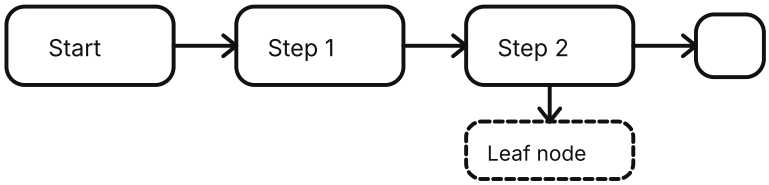
- **Polygons:** triangles, quadrilaterals, pentagon to octagon.
- **Curves:** circle, semicircle, oval, moon, cloud, heart, wedge.
- **Arrows & lines:** straight, curved, S-curve, single / double / curved / return arrows.
- **Flow symbols:** rounded rectangle, diamond, cylinder, document, data / IO, callout.

Edit After Inserting

- Select to move, scale and rotate with the corner handles.
- Change colour or thickness and press the shape again to restyle new shapes.
- Shapes snap like ink when drawn along instruments.
- Delete with the on-object control or the Section Eraser.

20. Flowchart & Process Tree Builder

Open **Flowchart** in the palette, set up the diagram in the dialog and press **Insert Flowchart on Page**. Blocks, arrows and text all use crisp vector shapes.



Dialog option	Choices
1. Direction	Horizontal Process Flow (left → right) or Vertical Tree (top → down).
2. Structure	**Process Flow** (sequential steps), **Tree Hierarchy** (root and leaves) or **Single Node** (extend manually).
3. Steps / Columns	Use – and + to set 1 or more steps (default 3).
4. Block Theme & Colour	Blue, Green, Purple, Orange, Slate or Custom colour for the blocks.

21. Editing in the Page: Floating Flowchart HUD

Flowchart HUD Buttons

- **Add Step** – adds a connected next block.
- **Leaf Node** – adds a subordinate branch under the selected block.
- **Text** – opens "Edit Flowchart Block Text".
- **Delete** – removes the selected block.
- **↔ / ↕ Dir** – toggles horizontal / vertical direction.
- **×** – closes the toolbar.

On-Block Controls & Gestures

- Selecting a block shows round handles: Add Connected Next Node, Add Subordinate / Leaf Node, Edit Block Text, Delete Block and Colour Theme.
- **Double-click** a block to rename it quickly.
- **Pull / push** the connector handles to grow or fold branches.
- Each block can have its own colour theme; a status line shows the active block.

22. Examples & Best Practices

Use cases

- Algorithm flowcharts and decision trees.
- Chapter mind-maps and family trees.
- Chemistry and biology process chains.
- Project stages and exam revision maps.

Best practices

- Keep block text short (2–4 words).
- Use Process Flow for steps and Tree for hierarchy.
- Pick one colour theme per diagram for printing.
- Use Select to move whole blocks; Undo reverses edits.

Combine tools

Flowchart blocks sit on the same page as ink, shapes and instruments. Annotate with the Pencil, highlight key blocks with the Highlighter and export the page to PDF, all without leaving the board.

23. Lazy-Loading Engine: Open Huge Files Without Crashing

Importing a large book, slide deck or workbook no longer loads everything into memory. KryloCanvas reads only what it needs, keeps just a few pages “awake”, and puts the rest to sleep. This keeps tablets and low-memory devices fast and stable.



How the PDF Engine Works

- **Byte-range loading:** only the PDF’s structure and the pages you need are read, not the whole file.
- **Page shells:** the first 20 pages appear at once; remaining shells are created in background batches.
- **≤ 5 awake pages:** about 3 visible plus one buffer on each side are rendered; others sleep.
- **Sleeping pages** shrink their canvas to 1 × 1 px to free memory, then restore when scrolled back.
- **Page cache:** up to 40 compressed page images (30 on low-memory devices) are kept for quick return.
- **Pixel cap:** each rendered page is limited to about 3.4 MP (2.3 MP on low-memory devices).

Safe Export of Big Documents

- PDF export loads and writes pages **one at a time**, then releases each page.
- Your drawings are kept on top of the original page, in vector quality.
- Choose **PDF (Current Page)** for a quick single sheet.
- The Rename Export dialog lets you add a name in front of the file name.

24. Word, Excel & PowerPoint Import

File type	How it is converted	Notes
Word (.docx)	Read by the built-in converter and split into 1200 × 1600 pages.	Works with MS Word and Google Docs downloads. Large embedded pictures are slimmed.
Excel (.xlsx, .xls, .csv)	Each sheet is cut into pages of **34 rows × 7 columns**.	Wide sheets continue on extra pages in reading order.
PowerPoint (.pptx)	Slides are rendered one by one as board pages.	Only the file’s index is read first; media over 3 MB is replaced with a placeholder.
PDF	Lazy byte-range pipeline described above.	Best choice for textbooks and scanned notes.

Support libraries (zip reader, Word and Excel converters) are fetched only when you first import that file type, so the app starts quickly and offline use of the board itself is unaffected.

25. Limits, Tips & Troubleshooting

Tips for the Smoothest Import

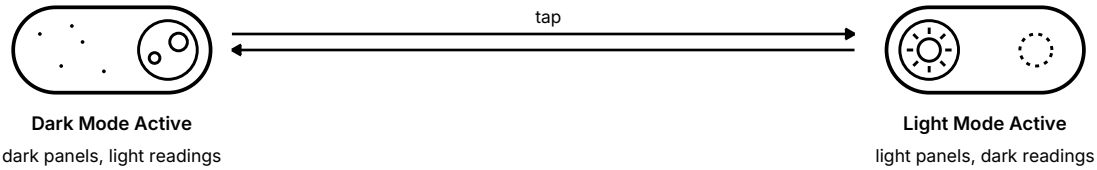
- Import one big file at a time and wait for pages to appear.
- Scroll steadily; pages wake as they come into view.
- Close other heavy browser tabs on tablets.
- Annotate first, export once: Export reads page after page.

If Something Looks Off

- A blank sleeping page fills in within a moment of scrolling.
- Very large images or media may be shown as placeholders.
- Use **Refresh** only if needed; it asks for confirmation.
- Re-import the file if a page fails to render.

25A. Light Mode & Dark Mode Switch

The **day / night pill** in the top bar (beside the title) switches the whole workspace between **Dark Mode** and **Light Mode** with one tap. The change is instant and smooth, and every tool keeps its settings.



What the switch changes

- **Header and palette** – top bar, tool palette and page area change to dark or light surfaces.
- **Page theme** – switching to Light turns a Black page White; switching to Dark turns a White page Black.
- **Default ink** – the starting stroke colour adapts so new writing is always readable.
- **Instruments** – ruler, protractor and set-square readings are white in Dark Mode and dark in Light Mode.
- A short message confirms the mode that is now active.

Smooth, adaptive interface

- **Welcome screen** – tap **Let's go** to enter the board.
- **Menu button** (three bars, top left) – opens Copyright & Credits.
- **Guide button** – **Read and access the instruction guide** sits in the header; its text stays white in both modes.
- **One-line header** – the top bar compacts step by step on tablets and phones instead of wrapping.
- **Visible palette scrollbar** on phones, tablets and short screens, so no tool looks cut off.

25B. Fountain Pen

Fountain Pen sits right below **Pen** in the palette. It writes smooth, slightly bolder ink with rounded ends – ideal for headings, titles and neat handwriting. It follows the universal colour and thickness like every other tool.



Fountain Pen stroke – round caps and joins, width = 1.3 × thickness

Tool	Stroke width	Look and best use
Pen	= thickness	Continuous bold vector ink for derivations.
Fountain Pen	= max(1, 1.3 × thickness)	Slightly bolder, round-ended ink for headings.
Pencil	= max(1, thickness ÷ 2)	Fine graphite line for construction work.
Highlighter	= 4 × thickness, 40 % opacity	Translucent wash over text and formulas.

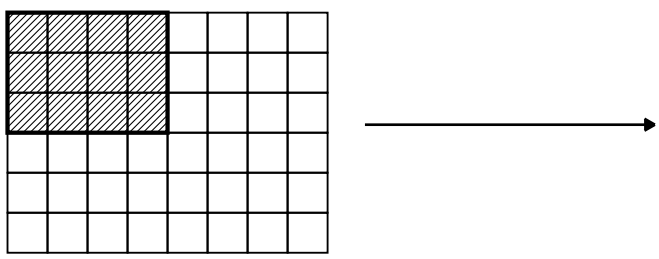
25C. Eraser Suite – Two Groups

The palette now holds **two eraser buttons**. Tap a button to open its pop-up, then pick the eraser you want. Instruments, flowchart blocks and memos are skipped by the Normal and Pixel Lasso erasers.

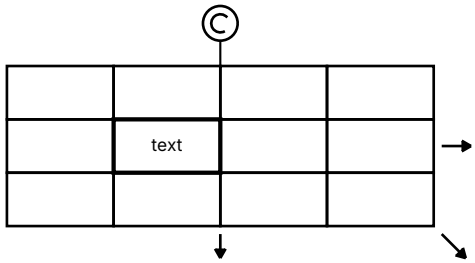
Eraser	Group	How you use it
Section Eraser	Specialized Erasers	Drag a dashed box; every object inside is deleted in one step (one Undo restores it).
Region Eraser	Specialized Erasers	Scribble over a line; only the part it touches is removed (fix a sign or an exponent).
Normal Eraser	Eraser	A circle follows your touch and rubs out what it covers. Five size presets (8, 16, 28, 44, 70 px) plus a slider.
Pixel Lasso Eraser	Eraser	A white streak erases only its own path. Size slider 2–80 px (default 12 px).

25D. Table Tool – Insert & Place

Tap **Tables** (below Memo in the palette) to build a table on the active page. Tables are vector objects: they move, rotate, stretch and print crisply, and every edit can be undone.



Grid selector: hover, then tap – 3 × 4



Rotate (top arrow) · stretch (edge arrows)

Insert dialog

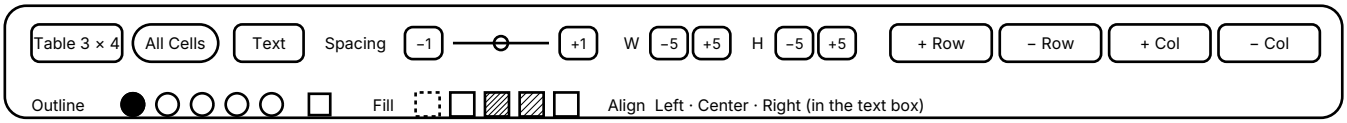
- **Word-style grid selector** – move over the grid to highlight rows × columns, then tap to insert.
- **Numeric option** – type exact Rows and Columns (up to 100 × 100) or use the – 1 / + 1 buttons.
- A live label shows the chosen size, e.g. 3 × 4.

On the page

- **Move** – drag the table with Select.
- **Rotate** – use the circular arrow above the table.
- **Stretch / compress** – drag the edge and corner arrows.
- **Type** – double-click (or double-tap) a cell to write in it.

25E. Floating Table Toolbar

Selecting a table opens a toolbar. A badge shows the table size and what is selected – **All Cells**, or the cell, row or column you tapped. Edits apply to that selection.



Control	What it does
Text	Opens the cell text box (or double-click a cell). Choose Left, Center or Right alignment, then Save.
Spacing	−1 / slider / +1 sets the gap between cells, 0 to 30 px.
W and H	−5 / +5 changes the width and height of the selected cell, column or row.
+ Row / – Row	Adds a row below the selection or deletes the selected row.
+ Col / – Col	Adds a column to the right of the selection or deletes the selected column.
Outline	Border colour: black, white, blue, green, red or any custom colour.
Fill	Cell fill (no fill, white, quick swatches or custom) for the selected cell or the whole table.

25F. Table Examples & Best Practices

Use cases

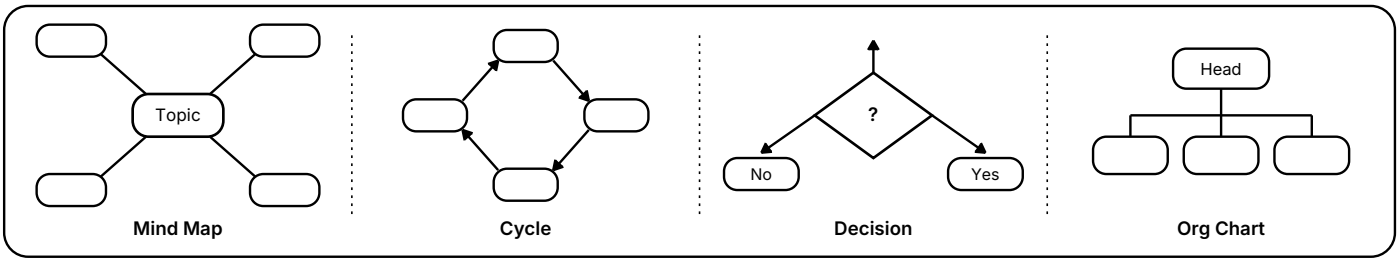
- Timetables, schedules and mark sheets.
- Comparison charts and data tables.
- Truth tables, periodic-table extracts and formula tables.
- Question grids for quizzes and revision.

Best practices

- Choose a size close to what you need; add rows later with + Row.
- Tap one cell, row or column before changing fill or size.
- Keep **All Cells** selected to restyle the whole table at once.
- Use Outline black and Fill none for ink-saving prints.

25G. New Flowchart Layouts

Besides **Process Flow**, **Tree Hierarchy** and **Single Node** (Section 20), the Flowchart dialog now offers four more layouts. Pick the layout, set the number of steps with – and +, choose a block colour and press **Insert Flowchart on Page**.



Layout	Structure	Best for
Mind Map	A central topic with branches in every direction.	Chapter summaries, brainstorming, revision maps.
Cycle	Blocks joined in a closed loop of arrows.	Water cycle, life cycles, repeating processes.
Decision	Question diamonds with Yes / No branches.	Algorithms, choices and troubleshooting.
Org Chart	A head block over rows of subordinates.	Family trees, school and team structures.

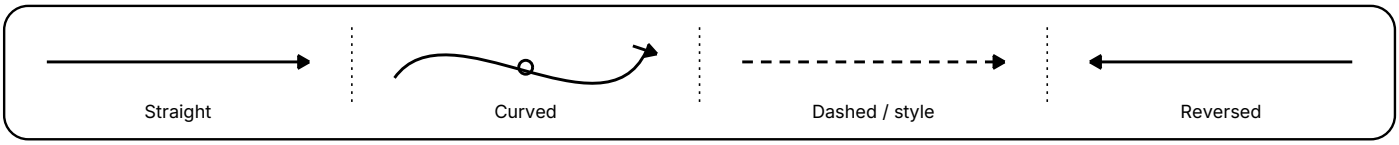
25H. Branch Tools in the Flowchart Toolbar

Select a block to use the extra buttons on the floating Flowchart toolbar. Every block can also be renamed with its green pencil handle, or by double-clicking it.

Button	What it does
Leaf Node / Sibling	Adds a child below the selected block, or a sibling beside it.
×2 ×3 ×5	Fan sub-trees: adds two, three or five child blocks at once.
Fold	Folds or unfolds everything below the selected block.
Tidy	Re-arranges the block's whole sub-tree neatly.
Colour themes	Blue, Green, Purple, Orange, Slate or a custom colour with a glowing halo.

25I. Full Arrow Tools

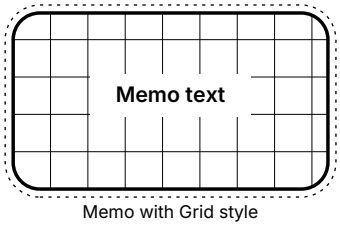
Arrows stay glued to their blocks while you move them. Select an arrow and use the toolbar row that appears; drag its green curvy-arrow handle to bend it by hand.



Arrow button	What it does
Straighten	Makes the arrow perfectly straight again.
Reverse	Swaps the start and end so the arrow points the other way.
Flip / Bend 45°	Moves the curve to the other side, or turns the bend by 45°.
Curve slider	Sets how strongly the arrow bends.
Style / Head	Changes the line style (solid, dashed) and the arrowhead shape.

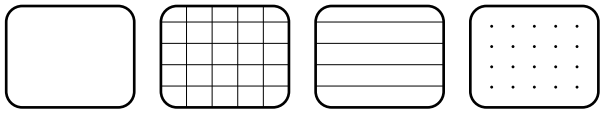
25J. Memo – Sticky Notes

Memo (below Flowchart in the palette) drops a rounded note on the active page, with a soft glowing halo so it stands out from the drawing. Use memos for reminders, definitions and key formulas.



Memo with Grid style

Page memo styles



Insert dialog

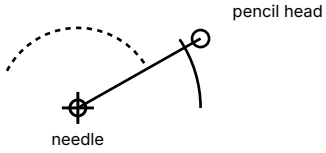
- **Colour ring** – tap a colour for the note.
- **Custom** – opens the full wheel with spectrum, white–black, R G B sliders and a HEX box.
- **Page memo style** – Blank, Grid, Lines or Dots.
- Press **Insert Memo** to place the note on the page.

Memo toolbar

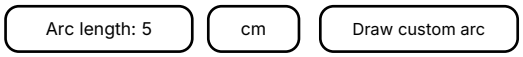
- **Text** – write or edit the note (also the pencil handle on the note).
- **Colour** – change colour or page style.
- **Style** – cycles Blank / Grid / Lines / Dots.
- **Delete** removes the memo; **X** closes the toolbar.

25K. Compass Additions

The Precision Compass of Section 17 keeps every feature and gains two more ways to work.



Custom Arc



Exact arc lengths in mm, cm or inch

Fix or detach the needle

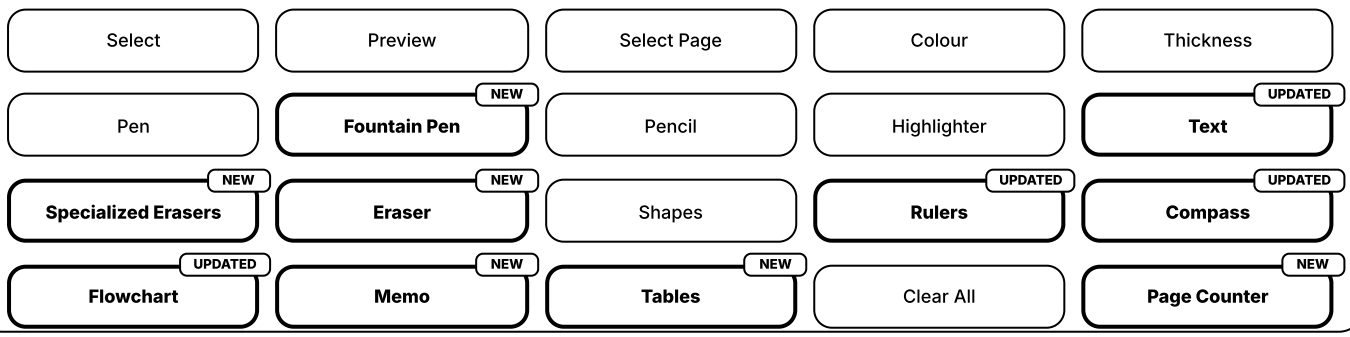
- Hold the red needle tip to **Fix** it or **Detach** it.
- Then drag the pencil head to swing the arm, or to draw arcs freely.

Custom Arc

- Choose **Custom Arc**, type an exact arc length and unit (mm / cm / inch) and draw it.
- Circle and Arc (90°) work as before.

25L. Updated Tool Palette

The palette keeps this fixed top-to-bottom order. Tools marked **NEW** were added and **UPDATED** were improved in this version.

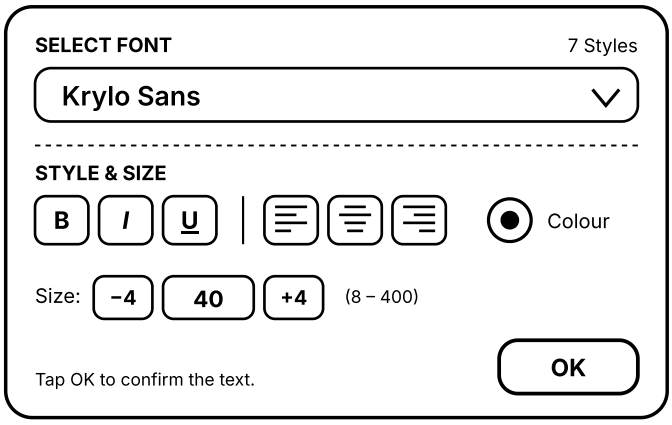


Combine tools

- Mix a Fountain Pen heading, a table of data, a mind map and a memo on one page, then export it as PDF – everything stays vector and sharp.

25M. Text Studio – Fonts, Style, Size & Colour

The Text tool now opens a dedicated **Text Studio** popup. Choose a font, switch on Bold, Italic or Underline, set an exact size and pick any colour – then tap **OK**.



How it works

- Choose **Text**, tap a page, type and press **OK**.
- **Select Font** opens a list where every font name is shown in its own typeface, so you can preview before choosing.
- **B / I / U** – bold, italic and underline switch on and off independently.
- **Align** – left, centre or right for multi-line text.
- **Size** – type any value from 8 to 400, or step with −4 / +4 (default 40).
- **Colour** – full spectrum picker, the same as the pen toolbar.
- Tap existing text with the Text tool to reopen its font, style, size and colour.

Everything stays vector

- Text can be moved, scaled and rotated with **Select**.
- Exports to PDF stay sharp at any zoom.
- Text prints crisply because it is stored as vector text, not as a picture.

Good to know

- The chosen font, style, size and colour are remembered for the next text.
- Enter starts a new line; an empty text box is removed.

25N. The Krylo Font Collection – 7 Styles

Each style below is written in its own typeface, exactly as it appears in the font list.

#	Font & sample	Look	Best used for
1	Krylo Sans The quick brown fox 0123	**Modern Clean**	Default style. Neat everyday writing, notes and labels.
2	**Krylo Sans Plus** **The quick brown fox 0123**	**Rounded Display**	Soft, rounded letters for titles and friendly worksheets.
3	Krylo Sans Premia The quick brown fox 0123	**Classic Serif**	Elegant high-contrast serif for headings and definitions.
4	KRYLO SANS DOTTED THE QUICK BROWN FOX 0123	**Dot-Matrix**	Nothing-style dot-matrix look for labels and tech themes.
5	**Krylo Comic** **The quick brown fox 0123**	**Cartoon & Comic**	Playful cartoon style for kids' sheets and speech notes.
6	Krylo Mono The quick brown fox 0123	**Technical Monospace**	Equal-width letters for code, tables and formulas.
7	*Krylo Script* *The quick brown fox 0123*	**Handwritten Cursive**	Flowing handwriting for greetings and personal notes.

Style buttons work on every font

Bold *Italic* Underline ***Bold Italic Underline*** Mono Bold

25O. One Font Engine – Tables, Flowcharts & Memos

The same 7 Krylo fonts, Bold / Italic / Underline, alignment and text colour are available everywhere you type – not only with the Text tool.

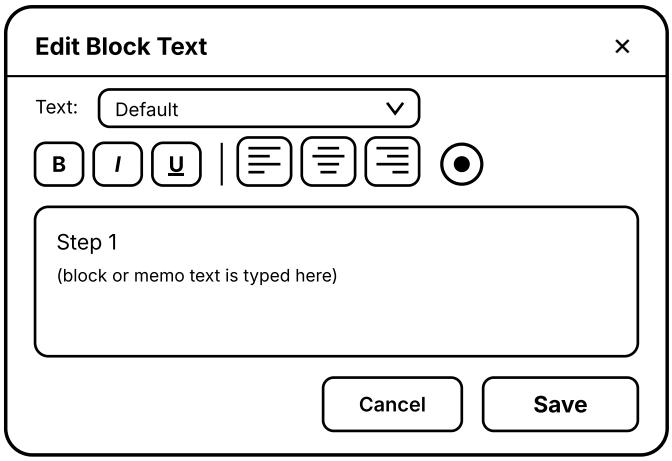


Table cells – a different font per cell, row or column

Krylo Sans	Name	Roll
Krylo Sans Plus	**Subject**	**Mark**
Krylo Sans Premia	*Maths*	*95*
Krylo Mono	Physics	88
Krylo Script	Chemistry	91

Flowchart blocks & memos

- Tap the green pencil (or double-click) to open **Edit Block Text**.
- Choose **Default** or any of the 7 fonts, then B / I / U, alignment and colour.
- Press **Save** to apply.

Where fonts work

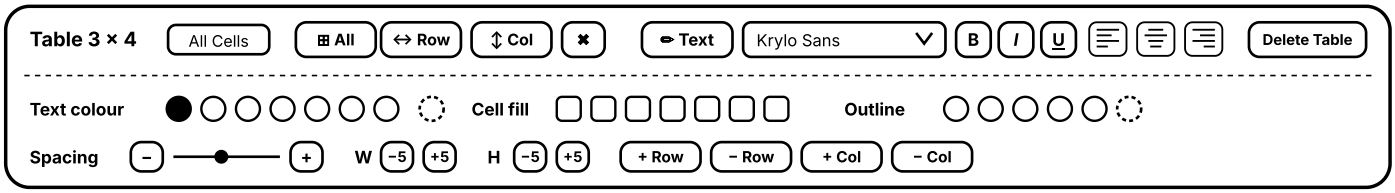
- **Text tool** – any text on the page.
- **Table cells** – font list in the table toolbar; applies to the selected cells.
- **Flowchart blocks** – each block can have its own font.
- **Memos** – sticky-note text uses the same dialog.

Writing in a table cell

- Double-click (double-tap) a cell, or press the **Text** button.
- Choose Left, Centre or Right, then **Save**.
- On a computer, **Ctrl + Enter** saves the cell text.
- Select a cell, row or column first so the font applies only there.

25P. Table Toolbar – Complete Reference

Section 25E introduced the table toolbar. These are the controls it also holds – the full strip is shown below.



Selecting cells

- **⊞ All** – every cell (the badge reads All Cells).
- **↔ Row** / **↕ Col** – select the row or column of the tapped cell.
- **✖** – clear the selection.
- Edits apply only to what is selected.

Colours

- **Text colour** – Black, White, Blue, Green, Red, Purple, Amber, or any custom colour.
- **Cell fill** – None, White, Soft Blue / Green / Yellow / Purple, Dark Slate or custom.
- **Outline** – Black, White, Blue, Green, Red or custom.

Size & spacing

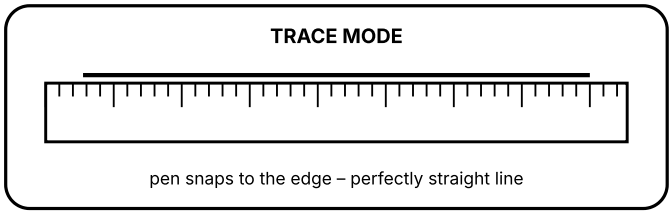
- **W** and **H** boxes accept exact pixel values, decimals included; −5 / +5 step quickly.
- **Spacing** slider and box set the gap between cells (0 – 30 px).

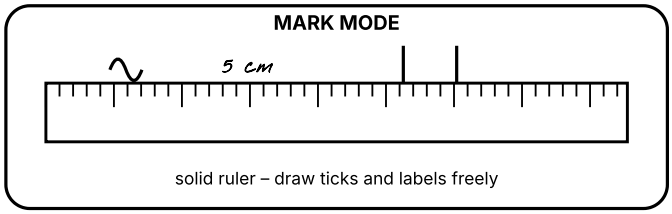
Rows, columns & removal

- + Row / − Row and + Col / − Col; a table always keeps at least 1 row and 1 column.
- **Delete Table** removes the whole table; **Close** hides the toolbar.

25Q. Instrument Modes – Trace or Mark

Open **Rulers** in the palette. Every instrument is offered in two modes, and there are now **four** instruments plus the compass.





Instrument	Trace mode (snapping)	Mark mode (solid)
Ruler	Straight lines along the scale edge	Freehand ticks, labels and notes
180° Protractor	Exact arcs and a straight base line	Degree ticks and angle notes
45° Set Square	Straight lines along the legs and hypotenuse	Freehand markings and labels
30°-60°-90° Set Square	Straight lines along the 30-60-90 edges	Freehand markings and labels

Instrument toolbar

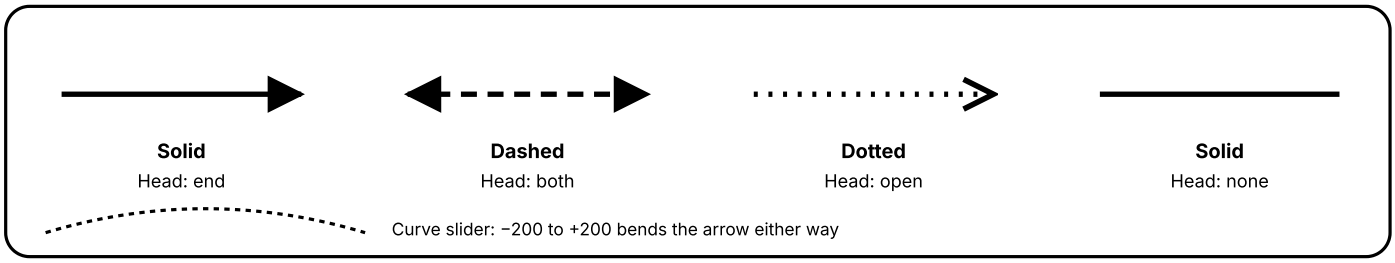
- Shows the instrument name and a badge – Trace Mode or Mark Mode.
- **Trace** / **Mark** buttons switch the mode on the placed instrument.
- **Remove** deletes it; ✖ closes the toolbar.

Choosing a mode

- Use **Trace** for exact lines, arcs and constructions.
- Use **Mark** to write degree labels, tick marks and notes around a solid instrument without the pen jumping to the edge.
- The **Precision Compass** button sits below the four instruments.

25R. Arrow Styling – Line, Head & Thickness

Select any flowchart arrow to reveal the full styling row (see 25I for Straighten, Reverse and Flip).



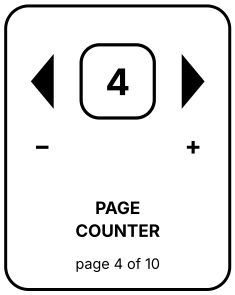
Style & head

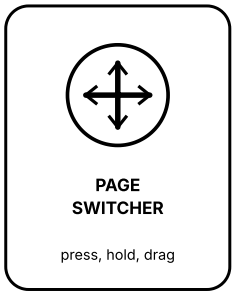
- **Style** button cycles Solid → Dashed → Dotted.
- **Head** button cycles End → Both → Open → None.

Thickness & bend

- **– / +** buttons make the arrow thinner or thicker.
- **Bend 45°** turns the curve; the slider sets how far it bends.

25S. Page Counter & Page Switcher





Page Counter (bottom of the palette)

- Shows the number of the current page.
- ◀ (−) goes to the previous page, ▶ (+) to the next, and scrolls it into view.
- The arrows grey out on the first and last page.

Page Switcher (on every page)

- Press, hold and drag the cross button to move a page to a new position.
- Needs at least 2 pages. **Esc** cancels the move.
- Stamp numbers (n / total) update automatically.

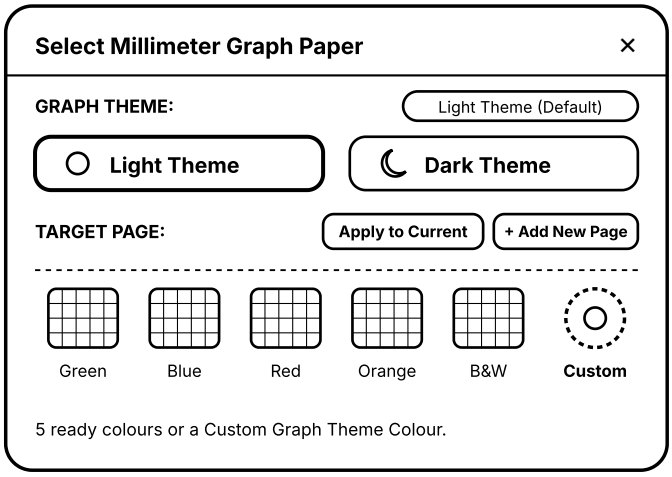
25T. New Features at a Glance

A quick checklist of everything added in sections 25M – 25V.

Feature	Where to find it	What it does
Text Studio	Text tool	Font, B / I / U, alignment, size 8 – 400 and colour in one popup
7 Krylo fonts	Text, tables, blocks, memos	Sans, Sans Plus, Sans Premia, Sans Dotted, Comic, Mono, Script
Table text styling	Table toolbar	Per-cell font, B / I / U, alignment, 7 text colours + custom
Table selection	Table toolbar	All, Row, Col and Clear; fill, outline, W / H, spacing; Delete Table
Block & memo text	Edit Block Text	Font, style, alignment and colour for flowchart blocks and memos
Instrument modes	Rulers popup	Trace (snaps) or Mark (solid) on ruler, protractor and two set squares
Arrow styling	Flowchart toolbar	Solid / dashed / dotted lines; end, both, open or no head; thickness
Graph paper themes	Millimeter Graph menu	Light or Dark graph paper, Custom colour, apply to current or new page
Compass modes	Compass pencil head	Draw arc freely or Move arm only, with angle and radius readout
Page Counter	Palette, bottom	Current page number with previous / next arrows
Page Switcher	Every page	Drag a page to a new position; numbering updates

25U. Graph Paper – Light, Dark & Custom Colour

The Millimeter Graph button now asks for a theme before it adds the paper, and you can mix your own colour.



Light & Dark graph paper

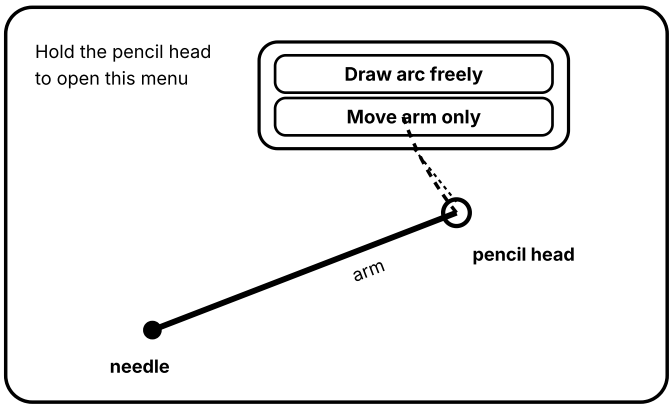
- Choose **Light Theme** (default) or **Dark Theme** before picking a colour.
- **Apply to Current** restyles the active graph page; **Add New Page** appends a fresh one.
- Page colour dots in the header do not change graph paper – set it here.

Custom Graph Theme Colour

- Tap the rainbow **Custom** circle.
- Pick Light or Dark mode, then use the spectrum, white–black bar, R G B or HEX.
- Lines, ticks and numbers adapt to the colour you choose.

25V. Compass – Draw Arc or Move Arm

Besides the needle menu of Section 25K, the pencil head has its own two-mode menu.



Draw arc freely

- Hold the pencil head, choose this mode, then sweep the head to draw an arc of any angle.
- A message shows the arc angle and the radius in cm.

Move arm only

- The arm swings around the needle without drawing – ideal for positioning before an arc.
- Switch back to Draw arc freely whenever you want to ink.

26. Canvas Themes, Page Styles & Graph Papers

KryloCanvas supports 3 contrast themes, a custom theme colour, 4 standard page styles and 5 millimeter engineering graph colours.

Themes & Contrast Inversion

- **White Theme:** clean background (`#ffffff`).
- **Black Theme:** dark mode (`#000000`); black strokes invert to white automatically.
- **Grey Theme:** neutral studio background (`#4a4a4a`).
- **Custom Theme Colour:** any colour from the spectrum; grid, lines and dots adapt.

Paper Styles & 5 Graph Papers

- **Blank, Standard Grid, Single Lines** (ruled) and **Dotted** grid.
- **5 Millimeter Graph colours:** Green, Blue, Red, Orange and B&W.
- **Rotate, Delete and Select Page** keep multi-page boards tidy.
- **Export:** PDF (all / current page) or image, with optional rename.

27. Official System Access & Software Credentials

KryloCanvas is freely accessible worldwide as a lightweight web application. To launch the full whiteboard studio:

Direct Web Deployment

krylocanvas.vercel.app

Point your smartphone, tablet or interactive flat panel camera at the QR code on the left to immediately load the whiteboard studio. Compatible with Chrome, Safari, Edge, Firefox, Android and iOS.

Website URL: **https://krylocanvas.vercel.app**

This guide: **krylocanvas.vercel.app/KryloCanvas-Instruction-Guide.pdf**

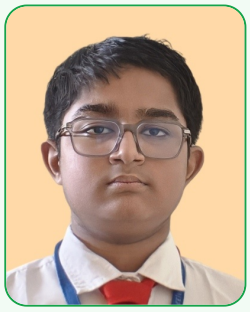
Software Architecture & System Engineering

About the Creator: Abhimat Chatterjee is an independent developer interested across multiple fields including **Robotics, STEM disciplines, algorithms, and interactive educational software**.

System Development: Driven to overcome the clunky latency and lack of precision in traditional digital boards, he conceptualized, designed, and coded from scratch as a lightweight, zero-dependency web application. It features a high-performance vector rendering pipeline, mathematical boundary-projection snapping for physical instruments, an extensible metric compass, dynamic flowchart trees, 43 geometric shapes and a lazy-loading engine for very large files.

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Abhimat Chatterjee
LEAD ARCHITECT & CREATOR



Photograph of the Creator

KryloCanvas (Web-WhiteBoard) • Built to empower teachers, inspire students, and elevate classroom instruction through mathematical rigor and digital simplicity.